Salamander Lemma on polycomplexes

Contents

1 Anstract

Questo paper generalizza il paper di Bergman a dimensioni di cocomplessi superiori.

2 Introduction

2.1 Notation

Adottiamo un sistema di indici interi per descrivere un n-cocomplesso. Data una categoria abeliana $\mathcal{A}$ consideriamo una famiglia di oggetti indicizzata su $\mathbb{Z}^n$

$$A = \{A_x\}_{x\in\mathbb{Z}^n}, \quad A_x \in \mathrm{Ob}(\mathcal{A})$$

Siano $e_i \in \mathbb{Z}^n$ l'i-esimo vettore della base canonica di $\mathbb{Z}^n$. Consideriamo n famiglie di differenziali fra gli oggetti

$$\partial^{(i)} = \{\partial_x^{(i)}\}_{x\in\mathbb{Z}^n},$$

per ogni direzione $i \in \{1, \cdots, n\}$ dove

$$\partial_x^{(i)} \in \mathrm{Hom}_{\mathcal{A}}(A_x, A_{x+e_i})$$

tali che:

1. ogni riga è un complesso di cocatene:

$$\partial_{x+e_i}^{(i)} \circ \partial_x^{(i)} = 0, \quad \forall x \in \mathbb{Z}^n,\ \forall i \in \{1, \cdots, n\}$$

2. commutatività del diagramma:

$$\partial_{x+e_i}^{(j)} \circ \partial_x^{(i)} = \partial_{x+e_j}^{(i)} \circ \partial_x^{(j)}, \quad \forall x \in \mathbb{Z}^n, i \neq j$$

Scriveremo $\pm$ per denotare il numero ± 1. Una coordinata $(\alpha_1, \cdots, \alpha_n)$ può essere scritta come $\alpha_1, \cdots, \alpha_n$. Una coordinata $(\alpha, \cdots, \alpha)$ verrà denotata semplicemente con α. Denotiamo pure sempre $A = A_0$.

Definition 2.1. *Se $x \leq y$ coordinata per coordinata, definiamo $\partial[x, y] \colon A_x \to A_y$ come la composizione lungo un qualsiasi cammino monotono da x a y. Per commutatività dei quadrati elementari, tale composizione è indipendente dal cammino scelto.*

2.2 First dimensional cases

For the one-dimensional case we have the cochain complex

$$A(-) \xrightarrow{\partial[-,0]} A \xrightarrow{\partial[0,+]} A(+)$$

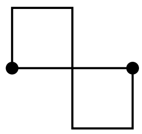
We can only define the classical comological object

$$H_-^+ \triangleq \frac{\ker \partial[0,+]}{\operatorname{im} \partial[-,0]}$$

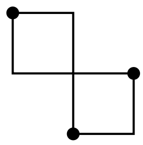
For the cochain bicomplexes we consider the following local commutative diagram

$$
\begin{tikzcd}
A(-) \arrow[r, "{\partial[(-),(0,-)]}"] \arrow[d, "{\partial[(-),(-,0)]}"] & A(0,-) \arrow[r, "{\partial[(0,-),+]}"] \arrow[d, "{\partial[(0,-),0]}"] & A(+) \arrow[d, "{\partial[+,(+,0)]}"] \\
A(-,0) \arrow[r, "{\partial[(-,0),0]}"] \arrow[d, "{\partial[(-,0),-]}"] & A \arrow[r, "{\partial[0,(+,0)]}"] \arrow[d, "{\partial[0,(0,+)]}"] & A(+,0) \arrow[d, "{\partial[(+,0),+]}"] \\
A(-) \arrow[r, "{\partial[-,(0,+)]}"'] & A(0,+) \arrow[r, "{\partial[(0,+),+]}"'] & A(+)
\end{tikzcd}
$$

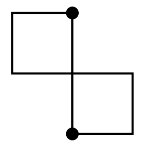
We define the directional homological objects, which are Bergman's objects, as follows:



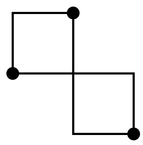
$$H^{+,0}_{-,0} \triangleq \frac{\ker \partial[0,(+,0)]}{\operatorname{im} \partial[(-,0),0]}$$



$$H^{(0,+),(+,0)}_{-} \triangleq \frac{\ker \partial[0,(+,0)] \cap \ker \partial[0,(0,+)]}{\operatorname{im} \partial[-,0]}$$



$$H^{+,0}_{-,0} \triangleq \frac{\ker \partial[0,(0,+)]}{\operatorname{im} \partial[(0,-),0]}$$

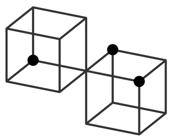


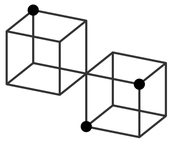
$$H^{+,-}_{(-,0),(0,-)} \triangleq \frac{\ker \partial[0,+]}{\operatorname{im} \partial[(-,0),0] + \operatorname{im} \partial[(0,-),0]}$$

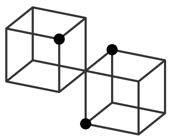
In thre dimensions we have the following commutative diagram:

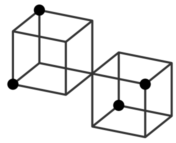
$$
\begin{tikzcd}
 & A(-,-,0) \arrow[rr] \arrow[dd] & & A(0,-,0) \arrow[dd] & & & \\
A(-) \arrow[ur] \arrow[rr] \arrow[dd] & & A(0,-,-) \arrow[ur] \arrow[dd] & & A(0,0,+) \arrow[rr] \arrow[dd] & & A(+,0,+) \arrow[dd] \\
 & A(-,0,0) \arrow[rr] & & A \arrow[ur] \arrow[rr] \arrow[dd] & & A(+,0,0) \arrow[ur] \arrow[dd] & \\
A(-,0,-) \arrow[ur] \arrow[rr] & & A(0,0,-) \arrow[ur] & & A(0,+,+) \arrow[rr] & & A(+) \\
 & & & A(0,+,0) \arrow[ur] \arrow[rr] & & A(+,+,0) \arrow[ur] &
\end{tikzcd}
$$

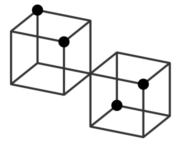
We have the following 18 comological objects:

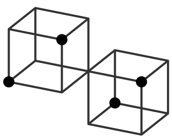

(1)

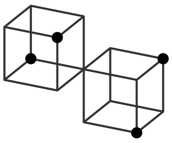

(2)

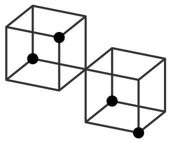

(3)

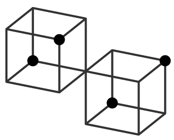

(4)

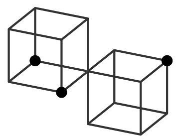

(5)

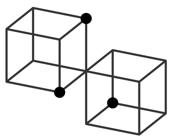

(6)

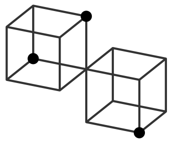

(7)

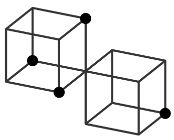

(8)

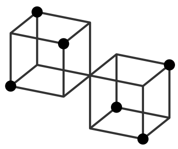

(9)

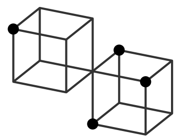

(10)

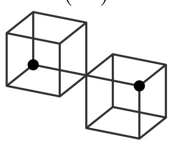

(11)

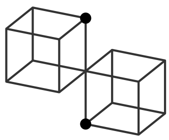

(12)

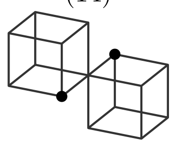

(13)


(14)


(15)


(16)


(17)


(18)

$$(1)\quad H^{(+,0,0),(0,0,+)}_{(-,0,0)} \triangleq \frac{\ker\partial[0,(+,0,0)]\cap\ker\partial[0,(0,0,+)]}{\operatorname{im}\partial[(-,0,0),0]}$$

$$(2)\quad H^{(+,0,0),(0,+,0)}_{(-,-,0)} \triangleq \frac{\ker\partial[0,(+,0,0)]\cap\ker\partial[0,(0,+,0)]}{\operatorname{im}\partial[(-,-,0),0]}$$

$$(3)\quad H^{(0,+,+),(0,0,+)}_{(0,-,-)} \triangleq \frac{\ker\partial[0,(0,+,0)]\cap\ker\partial[0,(0,0,+)]}{\operatorname{im}\partial[(0,-,-),0]}$$

$$(4)\quad H^{(+,0,0),(0,+,+)}_{(-,-,0),(-,0,-)} \triangleq \frac{\ker\partial[0,(+,0,0)]\cap\ker\partial[0,(0,+,+)]}{\operatorname{im}\partial[(-,-,0),0]+\operatorname{im}\partial[(-,0,-),0]}$$

$$(5)\quad H^{(+,0,0),(0,+,+)}_{(-,-,0),(0,-,-)} \triangleq \frac{\ker\partial[0,(+,0,0)]\cap\ker\partial[0,(0,+,+)]}{\operatorname{im}\partial[(-,-,0),0]+\operatorname{im}\partial[(0,-,-),0]}$$

$$(6)\quad H^{(+,0,0),(0,+,+)}_{(-,0,-),(0,-,-)} \triangleq \frac{\ker\partial[0,(+,0,0)]\cap\ker\partial[0,(0,+,+)]}{\operatorname{im}\partial[(-,0,-),0]+\operatorname{im}\partial[(0,-,-),0]}$$

$$(7)\quad H^{(+,+,0),(+,0,+)}_{(0,-,-),(-,0,0)} \triangleq \frac{\ker\partial[0,(+,+,0)]\cap\ker\partial[0,(+,0,+)]}{\operatorname{im}\partial[(0,-,-),0]+\operatorname{im}\partial[(-,0,0),0]}$$

$$(8)\quad H^{(+,+,0),(0,+,+)}_{(0,-,-),(-,0,0)} \triangleq \frac{\ker\partial[0,(+,+,0)]\cap\ker\partial[0,(0,+,+)]}{\operatorname{im}\partial[(0,-,-),0]+\operatorname{im}\partial[(-,0,0),0]}$$

$$(9)\quad H^{(0,+,+),(+,0,+)}_{(0,-,-),(-,0,0)} \triangleq \frac{\ker\partial[0,(0,+,+)]\cap\ker\partial[0,(+,0,+)]}{\operatorname{im}\partial[(0,-,-),0]+\operatorname{im}\partial[(-,0,0),0]}$$

$$(10)\quad H^{(+,0,+)}_{(0,0,-),(-,0,0)} \triangleq \frac{\ker\partial[0,(+,0,+)]}{\operatorname{im}\partial[(0,0,-),0]+\operatorname{im}\partial[(-,0,0),0]}$$

$$(11)\quad H^{(0,+,+)}_{(0,0,-),(0,-,0)} \triangleq \frac{\ker\partial[0,(0,+,+)]}{\operatorname{im}\partial[(0,0,-),0]+\operatorname{im}\partial[(0,-,0),0]}$$

$$(12)\quad H^{(+,+,0)}_{(-,0,0),(0,-,0)} \triangleq \frac{\ker\partial[0,(+,+,0)]}{\operatorname{im}\partial[(-,0,0),0]+\operatorname{im}\partial[(0,-,0),0]}$$

$$(13)\quad H^{+}_{(0,0,-),(-,0,0),(0,-,0)} \triangleq \frac{\ker\partial[0,+]}{\operatorname{im}\partial[(0,0,-),0]+\operatorname{im}\partial[(-,0,0),0]+\operatorname{im}\partial[(0,-,0),0]}$$

$$(14)\quad H^{(+,+,0),(0,+,+),(+,0,+)}_{(-,0,-),(0,-,-),(-,-,0)} \triangleq \frac{\ker\partial[0,(+,+,0)]\cap\ker\partial[0,(0,+,+)]\cap\ker\partial[0,(+,0,+)]}{\operatorname{im}\partial[(-,0,-),0]+\operatorname{im}\partial[(0,-,-),0]+\operatorname{im}\partial[(-,-,0),0]}$$

$$(15)\quad H^{(0,+,0),(+,0,0),(0,0,+)}_{-} \triangleq \frac{\ker\partial[0,(0,+,0)]\cap\ker\partial[0,(+,0,0)]\cap\ker\partial[0,(0,0,+)]}{\operatorname{im}\partial[-,0]}$$

$$(16)\quad H^{+,0,0}_{-,0,0} \triangleq \frac{\ker\partial[0,(+,0,0)]}{\operatorname{im}\partial[(-,0,0),0]}$$

$$(17)\quad H^{0,+,0}_{0,-,0} \triangleq \frac{\ker\partial[0,(0,+,0)]}{\operatorname{im}\partial[(0,-,0),0]}$$

$$(18)\quad H^{0,0,+}_{0,0,-} \triangleq \frac{\ker\partial[0,(0,0,+)]}{\operatorname{im}\partial[(0,0,-),0]}$$

2.3 Generalization of extramural maps

Definition 2.2. *Per ogni $x \in \mathbb{Z}^n$, e per ogni due famiglie finite $L \subseteq \mathbb{Z}^n_{\leq 0} \setminus \{0\}^n$, $U \subseteq \mathbb{N}^n \setminus \{0\}^n$ tali che abbia senso, poniamo*

$$I_L(x) \triangleq \sum_{\lambda \in L} \operatorname{im} \partial[x+\lambda, x], \quad K_U(x) \triangleq \bigcap_{\mu \in U} \ker \partial[x, x+\mu]$$

Quando vale

$$I_L(x) \subseteq K_u(x)$$

definiamo

$$H_L^U(x) \triangleq \frac{K_U(x)}{I_L(x)}$$

Concretamente, per gli oggetti comologici centrati su un oggetto centrale A_0 con struttura $L = \{l_i\}_{i=1}^h \subseteq \{-1, 0\}^n$ e $U = \{u_i\}_{i=1}^k \subseteq \{0, +1\}^n$ abbiamo

$$H_L^U = \frac{\bigcap_{i=1}^k \ker \partial[0, U_i]}{\sum_{i=1}^h \operatorname{im} \partial[L_i, 0]}$$

quando esiste. Ci stiamo quindi restringendo ad un intorno locale dell'oggetto centrale.

Nel paper di Bergman viene notato che il reticolo di oggetti comologici viene generato dai down-set. In generale questo numero combinatorico è il numero di Dedekind $M(n)$. Tuttavia, vengono scartati i casi banali del down-set vuoto e del down-set totale. Quindi in generale abbiamo

- $n = 1$: $M(1) - 2 = 1$
- $n = 2$: $M(2) - 2 = 4$
- $n = 3$: $M(3) - 2 = 18$
- $n = 4$: $M(4) - 2 = 166$
- $n = 5$: $M(5) - 2 = 7579$

Cominciamo generalizzando il concetto di *donor* e *receptor* come i gruppi comologici perfettamente diagonali che traversano l'oggetto centrale.

Definition 2.3. *The n-*receptor *centered at x is defined as*

$$R(x) \triangleq \frac{\bigcap_{i=1}^{n} \ker \partial[x, x+e_i]}{\operatorname{im} \partial[x-1, x]} = H_{-1}^{e_1, \cdots, e_n}(x)$$

Definition 2.4. *The n-*donor *centered at x is defined as*

$$D(x) \triangleq \frac{\ker \partial[x, x+1]}{\sum_{i=1}^{n} \operatorname{im} \partial[x-e_i, x]} = H^1_{-e_1, \cdots, -e_n}$$

Definition 2.5. *The i-*axial cohmology *centered at x is defined as*

$$H_i(x) \triangleq \frac{\ker \partial[x, x+e_i]}{\operatorname{im} \partial[x-e_i, x]} = H^{e_i}_{-e_i}$$

for $i \in \{1, \cdots, n\}$.

Cominciamo dimostrando esplicitamente il lemma 1.4 del paper originale per poi generalizzarlo.

Lemma 2.1. *Each arrow $f \colon A \to B$ in a double complex induces an arrow $A_\square \to {}^\square B$:*

Proof. Without loss of generality, let us consider the horizontal case:

$$\begin{tikzcd}
& \bullet \arrow[r] \arrow[d, "b"'] \arrow[dr, dashed, "p"] & \bullet \arrow[d] & \\
\bullet \arrow[r, "a"] & A \arrow[r, "f"] \arrow[d] \arrow[dr, dashed, "q"] & B \arrow[r, "d"] \arrow[d, "c"] & \bullet \\
& \bullet \arrow[r] & \bullet &
\end{tikzcd}$$

$$A_{\square} = \frac{\ker q}{\operatorname{im} a + \operatorname{im} b}, \quad {}^{\square}B = \frac{\ker c \cap \ker d}{\operatorname{im} p}$$

Let

$$K := \ker q, \quad M := \operatorname{im} a + \operatorname{im} b, \quad H := \ker c \cap \ker d, \quad N := \operatorname{im} p$$

we need to show that $f\colon A \to B$ induces a map $K/M \to H/N$. We need to ensure that the quotients are well-defined, meaning that $\operatorname{im} a + \operatorname{im} b \subseteq \ker q$ and $\operatorname{im} p \subseteq \ker c \cap \ker d$. Since the rows and columns are complexes and the squares commute, these inclusions hold.

- *f sends $\ker q$ into $\ker c \cap \ker d$*: let $\iota_K\colon \ker q \hookrightarrow A$ and $\iota_H\colon \ker c \cap \ker d \hookrightarrow B$ be the canonical inclusions. By definition of q and $\ker q$, $q\iota_K = cf\iota_K = 0$. By the properties of the complex, $df = 0$ and thus $df\iota_K = 0$. Thus, $f\iota_K\colon K \to B$ is nullified both by d and c. By the universal property of the intersection of kernels, meaning the pullback of $\ker c \cap \ker d$, there exists a unique morphism $\tilde{f}\colon K \to H$ such that $\iota_H \tilde{f} = f\iota_K$. Thus, f is restricted to $\tilde{f}\colon \ker q \to \ker c \cap \ker d$.
- *$\tilde{f}$ sends $\operatorname{im} a + \operatorname{im} b$ into $\operatorname{im} p$:* we can separately check $\operatorname{im} a$ and $\operatorname{im} b$. By the properties of the complex, $fa = 0$ meaning that fa factors through the suboject $\operatorname{im} p \hookrightarrow B$ via the zero morphism. By definition of p, $fb = p$ and thus the image of fb is contained in $\operatorname{im} p$. By the universal property of the sum of subobjects, $\operatorname{im} a + \operatorname{im} b$ is the smallest object of A containing both $\operatorname{im} a$ and $\operatorname{im} b$.

We now have the following diagram: $\begin{tikzcd} M \arrow[r, hook] \arrow[d] & K \arrow[r, two heads, "\pi_K"] \arrow[d, "\tilde{f}"'] & K/M \arrow[d, dashed, "\bar{f}"] \\ N \arrow[r, hook] & H \arrow[r, two heads, "\pi_H"'] & H/N \end{tikzcd}$ with $\pi_K\colon K \to K/M$ and $\pi_H\colon H \to H/N$ canonical projections.

Since $\pi_H \tilde{f}$ vanishes on M, by the universal property of the cokernel of $M \hookrightarrow K$, there exists a unique morphism $\bar{f}\colon K/M \to H/N$ such that $\bar{f}\pi_K = \pi_H \tilde{f}$. Thus, there exists a unique map

$$\bar{f}\colon \frac{\ker q}{\operatorname{im} a + \operatorname{im} b} \to \frac{\ker c \cap \ker d}{\operatorname{im} p}$$

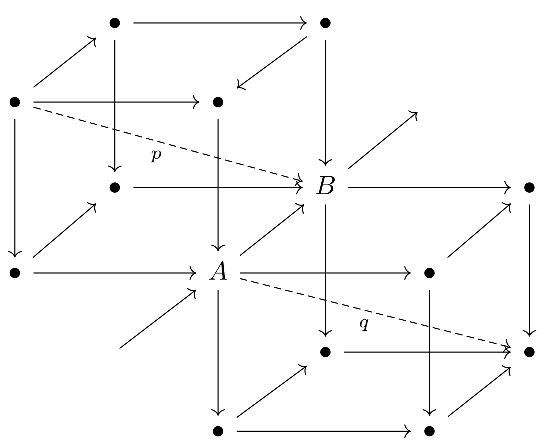
induced by f. $\square$

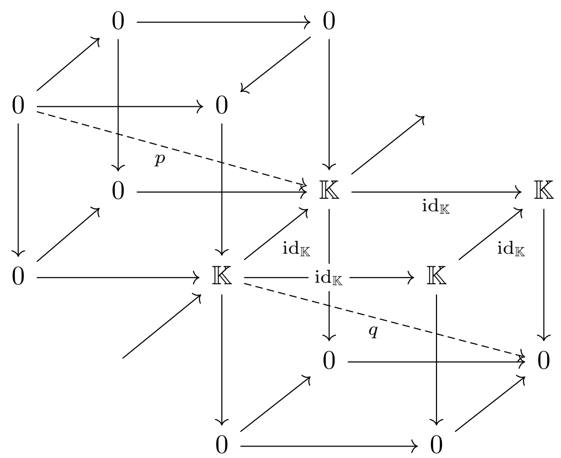
Written using our notation, this lemma states that for $n = 2$, a morphism $\partial[x, x+e_i]$ induces a morphism $D(x) \to R(x + e_i)$ for every $i \in \{1, 2\}$. The same lemma does not generalize for $n > 2$.

Proof. Consider the following counterexample: we consider $\mathbf{Vect}_{\mathbb{K}}$ for some field $\mathbb{K}$. We construct a 3-cocomplex where all the objects are zero except for

$$A = \mathbb{K}, \quad A_{1,0,0} = \mathbb{K}, \quad A_{1,0,1} = \mathbb{K}, \quad A_{0,0,1} = \mathbb{K} = B$$

All the maps are the null maps, except the maps between the latter objects which are the identity $\mathrm{id}_{\mathbb{K}}$.





The diagram commutes as the compositions of the non-trivial quare are still the identity. We will now compute the donor of A, $D(0)$ and the receiver of B, $D(e_3)$.

1. *donor of A:* we have

$$D(0) = \frac{\ker \partial[0,1]}{\operatorname{im} \partial[-e_1, 0] + \operatorname{im} \partial[-e_2, 0] + \operatorname{im} \partial[-e_3, 0]}$$

   In particular, every monotone path from coordinates 0 to 1 must go through the second dimension (downwards), which is the zero morphism. Thus, $\partial[0,1] = 0$. This means that every element of $\mathbb{K}$ is sent to 0 and thus $\ker \partial[0,1] = \mathbb{K}$. Furthermore, $A_{-e_1} = A_{-e_2} = A_{-e_3} = 0$ meaning that

$$\operatorname{im} \partial[-e_1, 0] + \operatorname{im} \partial[-e_2, 0] + \operatorname{im} \partial[-e_3, 0] = 0$$

   Thus, $D(0) = \mathbb{K}$.

2. *receiver of B:* we have

$$R(e_2) = \frac{\ker \partial[e_3, e_3 + e_1] \cap \ker \partial[e_3, e_3 + e_2] \cap \ker \partial[e_3, 2e_3]}{\operatorname{im} \partial[e_3 - 1, e_2]}$$

   For the kernels we have

$$\ker \partial[e_3, e_3 + e_1] \cap \ker \partial[e_3, e_3 + e_2] \cap \ker \partial[e_3, 2e_3] = \ker 0 \cap \ker 0 \cap \ker \mathrm{id}_{\mathbb{K}} = 0$$

   Furtermore, $A_{e_3 - 1} = A_{-1,-1,0} = 0$, and thus $\operatorname{im} p = 0$. Thus, $R(e_3) = 0$.

Suppose that $f = \mathrm{id}_{\mathbb{K}} \colon A \to B$ induces a morphism $D(0) \to R(e_3)$. Then, f must send elements of the numerator of $D(0)$ into the numerator of $R(e_3)$, meaning

$$f(\ker \partial[0,1]) \subseteq \ker \partial[e_3, e_3 + e_1] \cap \ker \partial[e_3, e_3 + e_2] \cap \ker \partial[e_3, 2e_3]$$

However, $\ker \partial[0,1] = \mathbb{K}$ and $f = \mathrm{id}_{\mathbb{K}} \colon A_0 \to A_{e_3}$ meaning that f doe snot factor through the zero subobject $0 \hookrightarrow A_{e_3}$. In particular the arrow $f \colon A_0 \to A_{e_3}$ does not restrict to an arrow

$$\ker \partial[0,1] \to \ker \partial[e_3, e_3 + e_1] \cap \ker \partial[e_3, e_3 + e_2] \cap \ker \partial[e_3, 2e_3]$$

Thus, f does not induce such a map. $\square$

Se provassimo a dimostrare il lemma falso per $n = 3$, l'unica inclusione che funziona sempre, sia per i nuclei che per le immagini, è quella nella direzione del morfismo iniziale. Considerando la definizione delle mappe diagonali, possiamo solamente dire l'ultima delle tre mappe composte viene annullata dal nucleo della mappa diagonale, ma ci servirebbe che pure le doppie composizioni vengano annullate, cioè ci servirebbe che l'oggetto comologico avesse intersezioni di nuclei di composizioni di due mappe, non solo una. Lo stesso ragionamento vale per il denominatore con le immagini. Notiamo che un tale oggetto è precisamente l'oggetto numerato (14), che è una sorta di middle object, sia donor che receiver. Allora, l'oggetto (13) si immerge naturalmente nell'oggetto (14), non il (15) come dimostratosi, mentre l'oggetto (14) si immerge naturalmente nell'oggetto (15). Possiamo quindi estendere il lemma originale aggiungendo un passaggio intermedio.

Definition 2.6. *The* 3-middle object *centered at x is defined as*

$$M(x) \triangleq \frac{\displaystyle\bigcap_{\substack{i,j=1\\ i\neq j}}^{3} \ker \partial[x, x+e_i+e_j]}{\displaystyle\sum_{\substack{i,j=1\\ i\neq j}}^{3} \operatorname{im} \partial[x-e_i-e_j, x]} = XXXXX$$

O più in generale

Definition 2.7. *The* n-middle object *centered at x is defined as*

$$M(x) \triangleq \frac{\displaystyle\bigcap_{\substack{k_1=1,\cdots,k_{n-1}=1\\ k_1\neq\cdots\neq k_{n-1}}}^{n} \ker \partial \left[x, x+\sum_{j=1}^{n-1} e_{k_j} \right]}{\displaystyle\sum_{\substack{k_1=1,\cdots,k_{n-1}=1\\ k_1\neq\cdots\neq k_{n-1}}}^{n} \operatorname{im} \partial \left[x-\sum_{j=1}^{n-1} e_{k_j}, x \right]} = XXXXX$$

Abbiamo allora il seguente lemma.

Lemma 2.2. *Each pair of arrows in a triple complex*

$$A \xrightarrow{f} B \xrightarrow{g} C$$

induces the arrows

$$D(A) \longrightarrow M(B) \longrightarrow R(C)$$

Proof. XXX $\square$

Da questa dimostrazione possiamo dedurre un criterio generale per quando un oggetto si immerge naturalmente in un altro oggetto comologico. Fissata una direzione e_i per ogni $i \in \{1, \cdots, n\}$ consideriamo due oggetti comologici arbitrari adiacenti nella direzione e_i:

$$A = \frac{X}{X}$$

3 Altro XXX

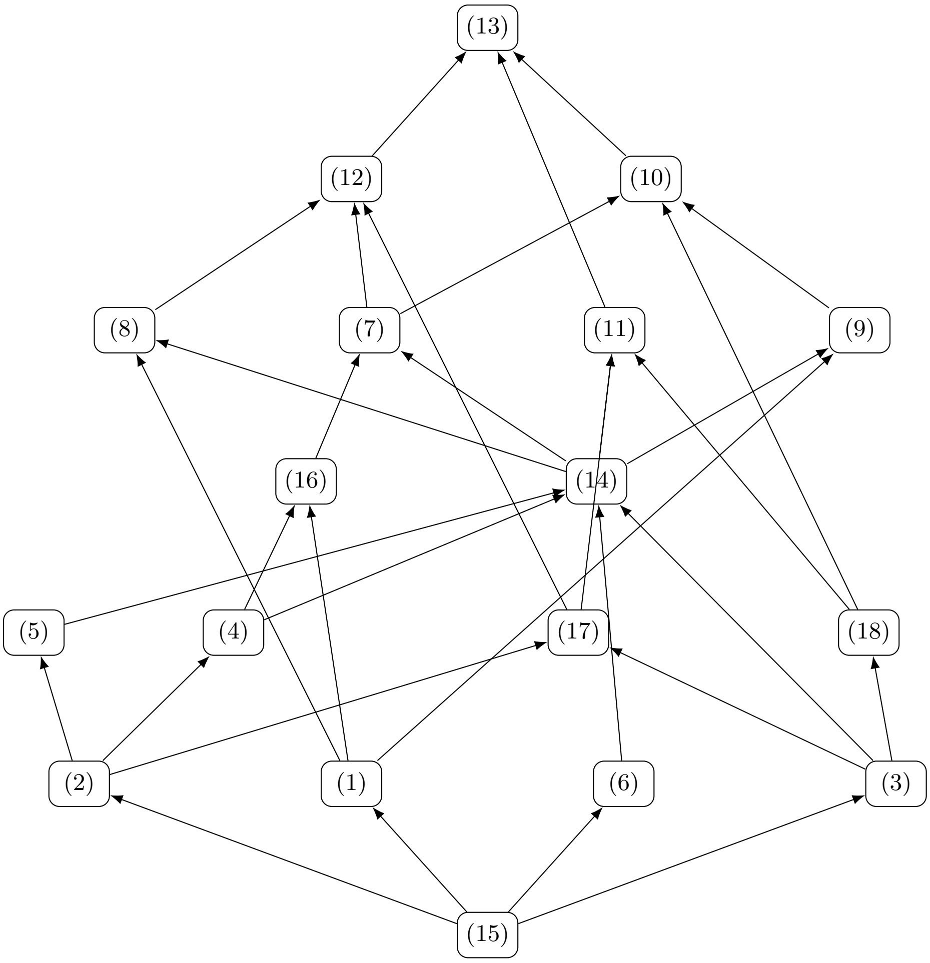
Caso verticale del diagramma del bicomplesso.

$$\begin{tikzcd}
 & U \arrow[d, "c"'] & \\
L \arrow[r, "d"'] \arrow[d, "b"'] & A \arrow[r, "e"] \arrow[d, "v"'] & C \arrow[d, "g"] \\
M \arrow[r, "k"'] & B \arrow[r, "h"'] \arrow[d, "m"'] & D \\
 & N &
\end{tikzcd}$$

$$D(A) = \frac{\ker(h \circ v)}{\operatorname{im} c + \operatorname{im} d} = \frac{\ker(g \circ e)}{\operatorname{im} c + \operatorname{im} d}, \qquad R(B) = \frac{\ker h \cap \ker m}{\operatorname{im}(v \circ d)} = \frac{\ker h \cap \ker m}{\operatorname{im}(k \circ b)}.$$

Ci piacerebbe dire che un morfismo A->B induce un morfiso D(A) -> D(B) questo già non è vero per n=3. Controesempio sugli spazi vettoriali. Che cosa ci servirebbe? Ci servirebbe un donor che ha l'intersezione dei 3 kernel a due step. Ma questo è l'elemento 14, infatti dati A->B->C abbiamo due mappe indotte D(A) -> M(B) -> R(C). Abbiamo quindi 3 * 2 = 6 casi, ma per la commutatività metà coincidono quindi 3.

E questo invece dovrebbe essere le intramural maps.


(13)
(12)
(10)
(8)
(7)
(11)
(9)
(16)
(14)
(5)
(4)
(17)
(18)
(2)
(1)
(6)
(3)
(15)